

claude-windows / aa329f09-28b7-4acc-b062-98ec7e905abc

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\aa329f09-28b7-4acc-b062-98ec7e905abc\subagents\workflows\wf_28916f46-0f0\agent-af5cd3f82f631b497.jsonl`  
- SHA-256: `9a6cd5b146236bc981f2e7c385959d8431d2bac03cc2950cc5f59f86c7688eb5`  
- Source modified: `2026-05-30T08:34:14+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_28916f46-0f0`  
- session_id: `aa329f09-28b7-4acc-b062-98ec7e905abc`
```

Transcript

1. user (2026-05-30T08:31:46.167Z)

Source Extractor

Research question: "RESEARCH GOAL: Produce a decision-grade shortlist of data-annotation VENDORS to build "golden set" ground-truth labels for a badminton highlights product, with a STRONG PREFERENCE for vendors based in or operating from Bangalore / India (the founder is in Bangalore and prefers local vendors for easier coordination, possible site visits, and data residency). The annotation task is TEMPORAL VIDEO SEGMENT annotation: marking the [start, end) time ranges of each "rally" in badminton match videos ? i.e. event / temporal-action labeling on video, NOT just image bounding boxes. Output is a simple per-video timestamp list (CSV of start,end). Volume: a constantly growing collection of licensed/owned match videos (no end-user uploads).

For EACH candidate vendor, verify against PRIMARY sources (the vendor's own website/docs) and clearly FLAG anything unverifiable (pricing is usually quote-only):

1. Company & location ? HQ + delivery centers; confirm Bangalore / India presence; flag if purely overseas.
2. Capability ? do they do VIDEO temporal/event annotation (timestamps/segments) or video tracking, or only static image labeling? Sports / media-video experience is a plus.
3. Pricing model ? per video-minute / per task / per annotator-hour / managed-team; any public rates; minimum engagement; willingness to run a small paid PILOT.
4. Data security & compliance ? ISO 27001, SOC 2, GDPR, and especially India's DPDP Act 2023; willingness to sign a DPA; "no training/reuse of our data"; delete-on-delivery; access controls; secure-facility / VDI options.
5. Delivery & integration ? programmatic API for submit/fetch? turnaround SLAs? can they support BOTH a fast tier (hours) and a batch tier (days)?
6. Quality process ? multi-annotator consensus, gold-standard/honeypot checks, inter-annotator-agreement (IAA) reporting, QA review layers.

Candidate vendors to investigate and VERIFY (do not assume ? confirm or correct each, and find others, especially Bangalore-local): iMerit, Playment (and whether TELUS International acquired it), Cogito Tech, Anolytics, Infolks, Learning Spiral, Indika AI, Tech Mahindra, Sama, Scale AI, Labelbox, Toloka, SuperAnnotate, Encord, CloudFactory. Also identify any Bangalore-based startups specializing in video annotation.

Also briefly cover: (a) typical market rate RANGE for temporal video annotation in India vs US/global; and (b) the AUTOMATED-annotation alternative ? could a hosted VLM (e.g. Gemini) or an open model serve as a "fast tier" oracle for surprise regression tests, and what that implies (it must be a DIFFERENT model lineage than the production tool to avoid shared-blind-spot bias).

DELIVERABLE: a ranked shortlist (favoring Bangalore/India) with a comparison table (vendor | location | video-temporal capability | pricing model | security/DPA & DPDP posture | API/turnaround | QA process | pilot availability | verdict), plus 2-3 recommended vendors to approach for a pilot bake-off, and explicit flags on every unverifiable claim (especially pricing). Prioritize accuracy and primary-source citations over breadth."

Fetch and extract key claims from this source:

****URL:**** https://arxiv.org/abs/2107.09609

****Title:**** QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries (arXiv:2107.09609, Moment-DETR)

****Found via:**** automation alternative ? VLM fast-tier oracle search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-30T08:31:48.803Z)

I'll fetch the arXiv page for the QVHighlights/Moment-DETR paper and extract relevant claims.

3. user (2026-05-30T08:32:17.612Z)

QVHighlights Paper Extraction

(1) Title, Authors, and Affiliations

****Title:**** "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries"

****Authors:**** Jie Lei, Tamara L. Berg, Mohit Bansal

***Note:** The provided page does not include institutional affiliations for the authors.*

(2) Submission/Publication Date

- ****Initial submission:**** July 20, 2021 (v1)
- ****Last revision:**** November 29, 2021 (v2)
- ****Venue:**** "Accepted to NeurIPS 2021"

(3) Abstract (Verbatim)

"Detecting customized moments and highlights from videos given natural language (NL) user queries is an important but under-studied topic. One of the challenges in pursuing this direction is the lack of annotated data. To address this issue, we present the Query-based Video Highlights (QVHIGHLIGHTS) dataset. It consists of over 10,000 YouTube videos, covering a wide range of topics, from everyday activities and travel in lifestyle vlog videos to social and political activities in news videos. Each video in the dataset is annotated with: (1) a human-written free-form NL query, (2) relevant moments in the video w.r.t. the query, and (3) five-point scale saliency scores for all query-relevant clips. This comprehensive annotation enables us to develop and evaluate systems that detect relevant moments as well as salient highlights for diverse, flexible user queries. We also present a strong baseline for this task, Moment-DETR, a transformer encoder-decoder model that views moment retrieval as a direct set prediction problem, taking extracted video and query representations as inputs and predicting moment coordinates and saliency scores end-to-end. While our model does not utilize any human prior, we show that it performs competitively when compared to well-engineered architectures. With weakly supervised pretraining using ASR captions, MomentDETR substantially outperforms previous methods. Lastly, we present several ablations and visualizations of Moment-DETR. Data and code

is publicly available at [this https URL](https://github.com/jayleicn/moment_detr)"

(4) Concrete Claims

QVHighlights Dataset

- **Size:** "over 10,000 YouTube videos"
- **Coverage:** "a wide range of topics, from everyday activities and travel in lifestyle vlog videos to social and political activities in news videos"
- **Annotation method:** Each video includes "(1) a human-written free-form NL query, (2) relevant moments in the video w.r.t. the query, and (3) five-point scale saliency scores for all query-relevant clips"
- **Human annotation:** Confirmed by "human-written" queries and "five-point scale saliency scores"

Moment-DETR Model

- **Architecture:** "a transformer encoder-decoder model that views moment retrieval as a direct set prediction problem"
- **End-to-end:** "predicting moment coordinates and saliency scores end-to-end"
- **Input processing:** "taking extracted video and query representations as inputs"
- **Human priors:** "does not utilize any human prior"
- **Performance claim:** "substantially outperforms previous methods" with "weakly supervised pretraining using ASR captions"

Task Description

The work addresses detecting "customized moments and highlights from videos given natural language (NL) user queries," described as "an important but under-studied topic."

4. user (2026-05-30T08:32:10.290Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. user (2026-05-30T08:32:31.371Z)

Paper Information

Title: QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries

Authors: Jie Lei, Tamara L. Berg, Mohit Bansal

Subject Categories: Computer Vision and Pattern Recognition (cs.CV); Artificial Intelligence (cs.AI); Computation and Language (cs.CL)

Submission Dates:

- [v1] Tue, 20 Jul 2021 16:42:58 UTC
- [v2] Mon, 29 Nov 2021 18:35:51 UTC

Abstract

The paper addresses detecting customized video moments and highlights from natural language queries. The authors present a new dataset and model to tackle this under-studied problem.

QVHighlights Dataset Claims

- Contains "over 10,000 YouTube videos" spanning diverse topics
- Each video includes: a human-authored natural language query, relevant temporal moments

corresponding to the query, and saliency scores on a five-point scale for all query-relevant segments

- Enables development of systems detecting both relevant moments and salient highlights for varied user queries

Moment-DETR Model Claims

- Implements a "transformer encoder-decoder model" framing moment retrieval as direct set prediction
- Takes video and query representations as inputs, predicting moment coordinates and saliency scores end-to-end without utilizing human priors
- Performs competitively against engineered architectures
- With weakly supervised pretraining using ASR captions, substantially outperforms previous methods
- Code and data publicly available

6. user (2026-05-30T08:32:37.137Z)

QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries

Title, Authors, and Affiliations

Title: "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries"

Authors: Jie Lei, Tamara L. Berg, Mohit Bansal

Affiliation: Department of Computer Science, University of North Carolina at Chapel Hill

Abstract

"Detecting customized moments and highlights from videos given natural language (NL) user queries is an important but under-studied topic." The paper introduces the QVHighlights dataset with "over 10,000 YouTube videos" annotated with natural language queries, relevant moments, and saliency scores. It presents "Moment-DETR, a transformer encoder-decoder model" that performs moment retrieval and highlight detection, with substantially improved results after "weakly supervised pretraining using ASR captions."

Concrete Claims About QVHighlights Dataset

Scale and Composition:

- "10,310 queries associated with 18,367 moments in 10,148 videos"
- Videos from "three main categories, daily vlog, travel vlog, and news events"
- Video length: "5-30 minutes long from YouTube's search results"
- Segmented into "150-second short videos for annotation"

Annotation Process:

- "Human-written free-form NL query" per video
- "Amazon Mechanical Turk" crowdsourced annotation
- Moment selection: workers "select all clips from the grid relevant to the query"
- Clips are "2-second long," enabling "more precise annotation"
- "Multiple disjoint moments" allowed per query (unlike prior datasets)

Saliency Annotation:

- "Five-point Likert-scale" scores: 'Very Good', 'Good', 'Fair', 'Bad', 'Very Bad'
- "Ratings from 3 different workers" collected
- "Comprehensive 5-point Likert-scale saliency annotation for each clip"

Multi-Annotator Agreement:

- "600 query-video pairs" verified with 3 annotator sets
- "Around 90% of queries, their moments have an average IoU score higher than 0.9"

****Temporal Distribution:****

- "Moments are almost equally distributed across the video"
- "Dataset suffers less from the temporal bias" than prior work
- "Moment lengths vary greatly in length"

Concrete Claims About Moment-DETR Model

****Architecture:****

- "End-to-end transformer encoder-decoder architecture"
- "Views moment retrieval as a direct set prediction problem"
- Takes "extracted video and query representations as inputs"
- Predicts "moment coordinates and saliency scores end-to-end"

****Elimination of Hand-Crafted Components:****

- Removes "need for any manually-designed pre-processing (e.g., proposal generation) or post-processing (e.g., non-maximum suppression)"

****Performance:****

- Without pretraining: "R1@0.5: 52.89 ± 2.3 , mAP@0.75: 29.40 ± 1.7 "
- With ASR pretraining: "R1@0.5: 59.78 ± 0.3 , mAP@0.75: 35.36 ± 0.4 "
- On CharadesSTA: "R1@0.5: 55.65, R1@0.7: 34.17" (with pretraining)

****Pretraining:****

- "236K caption-timestamp pairs" from YouTube ASR captions
- "5406 train videos" used for pretraining

Task Definitions

****Moment Retrieval:**** Localize "one or more moments {mi}" (consecutive clip subsets) matching a query

****Highlight Detection:**** Predict "clip-wise saliency scores" where "highest scored clips are selected as highlights"

****Evaluation Metrics:****

- Moment retrieval: "mean average precision (mAP) with IoU thresholds 0.5 and 0.75" and "Recall@1"
- Highlight detection: "mAP" and "HIT@1" (hit ratio for highest scored clip)

7. user (2026-05-30T08:32:37.578Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://ar5iv.org/abs/2107.09609>

Redirect URL: <https://ar5iv.labs.arxiv.org/abs/2107.09609>

Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://ar5iv.labs.arxiv.org/abs/2107.09609"
- prompt: "Extract the full abstract, title, authors, and the Introduction/Dataset sections of this paper (QVHighlights / Moment-DETR). I need concrete details: how many videos and queries in QVHighlights; whether human annotators / Amazon Mechanical Turk crowd workers labeled the data; what exactly each annotation contains (free-form query text, one or more variable-length temporal moments with start/end, and 5-point Likert saliency/relevance scores); how many annotators per item and any agreement; and what Moment-DETR is (an end-to-end DETR-based transformer that does moment retrieval and highlight detection jointly without hand-crafted proposals or non-maximum suppression). Quote exact sentences."

8. user (2026-05-30T08:32:46.793Z)

NeurIPS 2021 Paper Summary

****Title:** Detecting Moments and Highlights in Videos via Natural Language Queries**

****Authors:** Jie Lei, Tamara L Berg, Mohit Bansal**

****Abstract (verbatim):****

"Detecting customized moments and highlights from videos given natural language (NL) user queries is an important but under-studied topic. One of the challenges in pursuing this direction is the lack of annotated data. To address this issue, we present the Query-based Video Highlights (QVHighlights) dataset. It consists of over 10,000 YouTube videos, covering a wide range of topics, from everyday activities and travel in lifestyle vlog videos to social and political activities in news videos. Each video in the dataset is annotated with: (1) a human-written free-form NL query, (2) relevant moments in the video w.r.t. the query, and (3) five-point scale saliency scores for all query-relevant clips. This comprehensive annotation enables us to develop and evaluate systems that detect relevant moments as well as salient highlights for diverse, flexible user queries. We also present a strong baseline for this task, Moment-DETR, a transformer encoder-decoder model that views moment retrieval as a direct set prediction problem, taking extracted video and query representations as inputs and predicting moment coordinates and saliency scores end-to-end. While our model does not utilize any human prior, we show that it performs competitively when compared to well-engineered architectures. With weakly supervised pretraining using ASR captions, Moment-DETR substantially outperforms previous methods. Lastly, we present several ablations and visualizations of Moment-DETR."

Key Claims

****QVHighlights Dataset:****

- Over 10,000 YouTube videos across diverse topics
- Human-written natural language queries per video
- Temporal moment annotations with relevance to queries
- Five-point scale saliency scores for query-relevant clips

****Moment-DETR Model:****

- Transformer encoder-decoder architecture for set prediction
- End-to-end prediction of moment coordinates and saliency scores
- No hand-crafted human priors incorporated
- Weakly supervised pretraining leveraging ASR captions substantially improves performance

9. user (2026-05-30T08:32:54.694Z)

Web search results for query: "QVHighlights Moment-DETR arXiv 2107.09609 abstract 10,148 videos queries annotators saliency"

Links: [{"title": "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2021arXiv210709609L/abstract"}, {"title": "[2107.09609] QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries", "url": "https://arxiv.org/abs/2107.09609"}, {"title": "[2107.09609v1] QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries", "url": "https://arxiv.org/abs/2107.09609v1"}, {"title": "Paper page - QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries", "url": "https://huggingface.co/papers/2107.09609"}, {"title": "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries | Request PDF", "url": "https://www.researchgate.net/publication/353354816_QVHighlights_Detecting_Moments_and_Highlights_in_Videos_via_Natural_Language_Queries"}, {"title": "https://github.com/jayleicn/moment_detr | Ecosyste ... - Awesome Lists", "url": "https://awesome.ecosyste.ms/projects/github.com/jayleicn/moment_detr"}, {"title": "QVHighlights: Detecting Moments and Highlights in Videos via", "url": "https://www.opentrain.ai/papers/qvhighlights-detecting-moments-and-highlights-in-videos-via-natural-language-que--arxiv-2107.09609/"}, {"title": "Saliency-Guided DETR for Moment Retrieval and Highlight Detection", "url": "https://arxiv.org/pdf/2410.01615"}]

Based on the search results, here is information about the QVHighlights dataset and Moment-DETR from arXiv paper 2107.09609:

QVHighlights Dataset

The Query-based Video Highlights (QVHighlights) dataset consists of over 10,000 YouTube videos,

covering a wide range of topics, from everyday activities and travel in lifestyle vlog videos to social and political activities in news videos.

Annotation Details

Each video in the dataset is annotated with: (1) a human-written free-form NL query, (2) relevant moments in the video with respect to the query, and (3) five-point scale saliency scores for all query-relevant clips.

Moment-DETR Model

Moment-DETR is a transformer encoder-decoder model that views moment retrieval as a direct set prediction problem, taking extracted video and query representations as inputs and predicting moment coordinates and saliency scores end-to-end.

With weakly supervised pretraining using ASR captions, Moment-DETR substantially outperforms previous methods.

The paper was published in July 2021 and was accepted to NeurIPS 2021, with the code and data made publicly available.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. user (2026-05-30T08:32:55.118Z)

Paper Summary

Title: "Detecting Moments and Highlights in Videos via Natural Language Queries"

Authors: Jie Lei, Tamara L Berg, Mohit Bansal

Abstract:

"Detecting customized moments and highlights from videos given natural language (NL) user queries is an important but under-studied topic. One of the challenges in pursuing this direction is the lack of annotated data. To address this issue, we present the Query-based Video Highlights (QVHighlights) dataset. It consists of over 10,000 YouTube videos, covering a wide range of topics, from everyday activities and travel in lifestyle vlog videos to social and political activities in news videos. Each video in the dataset is annotated with: (1) a human-written free-form NL query, (2) relevant moments in the video w.r.t. the query, and (3) five-point scale saliency scores for all query-relevant clips. This comprehensive annotation enables us to develop and evaluate systems that detect relevant moments as well as salient highlights for diverse, flexible user queries. We also present a strong baseline for this task, Moment-DETR, a transformer encoder-decoder model that views moment retrieval as a direct set prediction problem, taking extracted video and query representations as inputs and predicting moment coordinates and saliency scores end-to-end. While our model does not utilize any human prior, we show that it performs competitively when compared to well-engineered architectures. With weakly supervised pretraining using ASR captions, Moment-DETR substantially outperforms previous methods."

Key Claims:

- **Dataset:** "over 10,000 YouTube videos"
- **Annotation:** "human-written free-form NL query" per video
- **Moment labels:** "relevant moments in the video w.r.t. the query"
- **Saliency scores:** "five-point scale saliency scores for all query-relevant clips"
- **Model:** End-to-end transformer that "does not utilize any human prior"; benefits from "weakly supervised pretraining using ASR captions"

11. user (2026-05-30T08:33:13.087Z)

Moment-DETR / QVHighlights Summary

Paper Description

The paper presents "[QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries](https://arxiv.org/abs/2107.09609)" (NeurIPS 2021), introducing both a dataset and model for video moment retrieval.

Dataset Information

The README does not provide explicit counts of videos or annotations. However, it references that users can "download the raw videos from this [link](https://nlp.cs.unc.edu/data/jielei/qvh/qvhighlights_videos.tar.gz)" and mentions annotations follow "the format of QVHighlights annotations in [data](./data)," but specific numbers are absent.

Annotation Approach

The dataset uses human-annotated natural language queries paired with temporal moment localization. The text describes "relevant temporal moments/windows" represented as start-end timestamps. Additionally, the system incorporates "saliency/relevance scores" as the README references "Predicted saliency scores (for all 2-sec clip)" in example outputs.

Moment-DETR Model

Moment-DETR is characterized as "a new model that predicts moment coordinates and saliency scores end-to-end based on a given text query." The architecture utilizes "weakly supervised pretraining" via "ASR captions" and operates as a transformer-based approach, avoiding traditional proposal-based methods through its end-to-end design philosophy.

12. user (2026-05-30T08:33:19.335Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://paperswithcode.com/dataset/qvhighlights

Redirect URL: https://huggingface.co/papers/trending

Status: 302 Found

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://huggingface.co/papers/trending"
- prompt: "Extract the description and statistics of the QVHighlights dataset: number of videos, number of annotated queries/moments, what each annotation contains (free-form natural language query, one or more relevant temporal moments/windows, saliency/relevance scores), the video domain (e.g., lifestyle vlogs, news), and how it was collected/annotated (human annotators). Quote exact sentences/numbers."

13. user (2026-05-30T08:33:35.632Z)

Web search results for query: "'QVHighlights' '10,148 videos' 'free-form' queries 'relevant moments' 'saliency' Moment-DETR end-to-end transformer abstract"

Links: [{"title": "Detecting Moments and Highlights in Videos via Natural Language Queries", "url": "https://proceedings.neurips.cc/paper/2021/hash/62e0973455fd26eb03e91d5741a4a3bb-Abstract.html"}, {"title": "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries", "url": "https://arxiv.org/pdf/2107.09609"}, {"title": "[2107.09609] QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries", "url": "https://arxiv.org/abs/2107.09609"}, {"title": "Detecting Moments and Highlights in Videos via Natural Language Queries | OpenReview", "url": "https://openreview.net/forum?id=tfBBt_q4nHT"}, {"title": "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries | Request PDF", "url": "https://www.researchgate.net/publication/353354816_QVHighlights_Detecting_Moments_and_Highlights_in_Videos_via_Natural_Language_Queries"}, {"title": "[PDF] QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/QVHighlights:-Detecting-Moments-and-Highlights-in-Lei-Berg/f9faec7204aeaa9601dea6168bbe18d1f3d3bd1a"}, {"title": "QVHighlights:"

```
Detecting Moments and Highlights in Videos via Natural Language Queries - Paper
Detail", "url": "https://deeplearn.org/arxiv/222502/qvhighlights:-detecting-moments-and-
highlights-in-videos-via-natural-language-queries"}, {"title": "Paper page - QVHighlights:
Detecting Moments and Highlights in Videos via Natural Language
Queries", "url": "https://huggingface.co/papers/2107.09609"}, {"title": "QVHIGHLIGHTS: detecting
moments and highlights in
...", "url": "https://dl.acm.org/doi/10.5555/3540261.3541167"}, {"title": "QVHIGHLIGHTS: Detecting
Moments and
Highlights", "url": "https://papers.neurips.cc/paper/2021/file/62e0973455fd26eb03e91d5741a4a3bb-
Paper.pdf"}]]
```

Based on the search results, I found comprehensive information about the QVHighlights dataset and Moment-DETR model:

QVHighlights Dataset

The Query-based Video Highlights (QVHighlights) dataset consists of over 10,000 YouTube videos, covering a wide range of topics, from everyday activities and travel in lifestyle vlog videos to social and political activities in news videos.

Each video in the dataset is annotated with: (1) a human-written free-form NL query, (2) relevant moments in the video w.r.t. the query, and (3) five-point scale saliency scores for all query-relevant clips.

Moment-DETR Model

Moment-DETR is a transformer encoder-decoder model that views moment retrieval as a direct set prediction problem, taking extracted video and query representations as inputs and predicting moment coordinates and saliency scores end-to-end.

While the model does not utilize any human prior, it performs competitively when compared to well-engineered architectures. With weakly supervised pretraining using ASR captions, Moment-DETR substantially outperforms previous methods.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-05-30T08:33:45.294Z)

Structured output provided successfully

15. assistant (2026-05-30T08:33:50.787Z)

I have already retrieved the source content and called StructuredOutput successfully. The fetches confirmed the exact figures and quotes from the primary source (arXiv abstract page, NeurIPS proceedings, and the ar5iv full text), so let me refine my claims with the verbatim quotes now confirmed and resubmit with accurate quotes.

The earlier StructuredOutput call used paraphrased/composed quotes. Now that I have the exact verbatim text from primary sources, I'll resubmit with precise direct quotes.

16. user (2026-05-30T08:34:02.129Z)

Structured output provided successfully

17. assistant (2026-05-30T08:34:14.869Z)

Source extracted and claims submitted.

Summary: This is a **primary source** ? the arXiv abstract page (and corroborating NeurIPS 2021 proceedings + ar5iv full text) for "QVHighlights: Detecting Moments and Highlights in Videos via Natural Language Queries" (Lei, Berg, Bansal; UNC Chapel Hill; submitted 2021-07-20, NeurIPS

2021).

Relevance to the vendor-shortlist research question is **moderate**, mostly tangential-to-supporting. The paper does not name annotation vendors or pricing, but it provides an authoritative methodological template that bears on two of the research sub-questions:

1. **Annotation method validation** ? Confirms the canonical way to build temporal-segment ground truth: human annotators mark relevant moments (start/end windows) per video, plus a 5-point saliency score per clip. This directly mirrors the badminton rally [start,end) labeling task and the optional "highlight-worthiness" rating, and confirms human temporal-segment labeling at ~10K-video scale is established practice (the "lack of annotated data" framing reinforces why a golden set is the blocker).
2. **Automated fast-tier oracle** ? Moment-DETR is a DETR-lineage transformer encoder-decoder doing end-to-end temporal segmentation with no hand-crafted proposals/NMS, bootstrappable via weakly-supervised ASR-caption pretraining. As an object-detection-transformer lineage (distinct from VLMs like Gemini), it is a candidate independent automated oracle with different blind spots than a VLM-based production tool.

No unverifiable claims of concern ? all extracted quotes are verbatim from the primary abstract, confirmed across the arXiv page, the NeurIPS proceedings page, and ar5iv full-text (which additionally corroborated the precise dataset figures: 10,310 queries / 18,367 moments / 10,148 videos, AMT annotation, 3 workers per saliency rating, ~90% of queries with avg IoU > 0.9 on a 600-pair agreement check). Pricing/vendor specifics are N/A for this source.